

A CAPSTONE PROJECT – I REPORT
ON
“KisanSangam – AI-Enabled Smart Agriculture Platform”

**Submitted in partial fulfillment of the requirements
for the award of the degree of**

**BACHELOR OF TECHNOLOGY (B.Tech.)
IN
COMPUTER SCIENCE AND ENGINEERING**

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CERTIFICATE

This is to certify that the Capstone Project – I Report entitled “**KisanSangam – AI-Enabled Smart Agriculture Platform**” submitted by Meet Barot, Enrollment No. 2301201110, and Manav Patel, Enrollment No. 2301201119, in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, has been successfully completed during the academic year 2026–2027.

This work has been carried out under the guidance and supervision of Ms. Miti Desai and is up to our satisfaction.

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DECLARATION

We hereby declare that the Capstone Project – I Report entitled **“KisanSangam – AI-Enabled Smart Agriculture and Farm Management Platform”** is a record of the original work carried out by us during the course of our Capstone Project – I.

This work has not been submitted, either in part or in full, to any other University or Institution for the award of any degree or any other academic qualification.

We further declare that all sources of information used in this report have been duly acknowledged and cited wherever necessary.

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ABSTRACT

Agriculture involves several interrelated decisions, including crop selection, cultivation planning, farm record management, market analysis, and understanding weather conditions. However, farmers often need to access information from multiple sources, making agricultural decision-making more difficult and time-consuming.

KisanSangam – AI-Enabled Smart Agriculture and Farm Management Platform is developed as an integrated digital platform intended to bring important agricultural information and farm management capabilities together in a single system. The platform consists of an introductory web application and a primary mobile application designed for farmers.

The current implementation focuses on five core functionalities: AI Crop Recommendation, Crop-Specific Farm Guide, Farm Management, Market Price Analysis, and Weather Reporting. The AI Crop Recommendation module uses machine learning techniques to process agricultural parameters and generate crop recommendations. The Crop-Specific Farm Guide provides relevant information associated with selected crops. The Farm Management module enables users to maintain digital records related to farms, crops, activities, expenses, and production. Market Price Analysis provides relevant agricultural market price information, while Weather Reporting provides current weather and forecast information.

The system integrates a mobile application, backend REST APIs, database services, AI/ML components, and external APIs. The user interface and user experience are designed to provide a centralized and accessible platform for farmers.

The project demonstrates how artificial intelligence, software engineering, data management, and digital agricultural services can be combined to develop an integrated agricultural decision-support platform. The modular architecture also provides a foundation for future enhancements such as soil-testing integration, advanced AI analytics, multilingual support, cloud scalability, and additional agricultural data integration.

Keywords: Artificial Intelligence, Smart Agriculture, Machine Learning, Crop Recommendation, Farm Management, Market Price Analysis, Weather Reporting, KisanSangam.

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CHAPTER 1: INTRODUCTION

1.1 BACKGROUND

Agriculture is one of the most important sectors of the Indian economy and provides livelihood opportunities to a significant portion of the population. Farming requires continuous decision-making related to crop selection, cultivation practices, farm activities, expenses, market conditions, and weather. Farmers often depend on traditional knowledge, local experience, agricultural experts, and multiple information sources to make these decisions.

With the rapid development of **Artificial Intelligence (AI), Machine Learning (ML), web technologies, and data-driven applications**, there is an opportunity to develop intelligent systems that can support farmers in making informed agricultural decisions. AI-based systems can analyze relevant agricultural parameters and provide recommendations, while digital platforms can simplify the process of accessing and managing farming-related information.

One of the important challenges faced by farmers is selecting an appropriate crop according to the available agricultural conditions. Crop selection can depend on several factors, and making an unsuitable choice may affect farm productivity and profitability. An intelligent crop recommendation system can assist farmers by analyzing relevant input parameters and suggesting suitable crops.

In addition to crop selection, farmers require information about **crop-specific cultivation practices**. Different crops have different requirements and farming procedures. Providing this information through a structured digital platform can make agricultural guidance more accessible.

Farmers also need to maintain information related to their farms, crops, activities, expenses, and production. Manual record keeping can make it difficult to organize and review this information. A digital **farm management** module can provide a systematic way of maintaining such records.

Market conditions are another important consideration in agriculture. Crop prices can vary across markets and over time. Providing market price information through a

centralized platform can help farmers access relevant market data when planning their agricultural activities.

Weather conditions also have an important relationship with farming operations. Information such as temperature, rainfall, humidity, and forecasts can be useful when planning agricultural activities. Therefore, providing **weather reporting** as part of the platform can give farmers convenient access to relevant weather information.

To address these requirements, **KisanSangam** is proposed as an **AI-enabled smart agriculture and farm management platform**. The system integrates five major functional areas:

1. **AI-Based Crop Recommendation**
2. **Crop-Specific Farm Guide**
3. **Farm Management**
4. **Market Price Analysis**
5. **Weather Reporting**

By bringing these functionalities together, KisanSangam aims to provide farmers with a centralized digital platform for accessing agricultural information, receiving AI-based recommendations, and managing important farm-related records.

1.2 PROBLEM STATEMENT

Farmers need to make several decisions throughout the agricultural cycle, including selecting suitable crops, understanding cultivation practices, managing farm activities, monitoring expenses, checking market prices, and considering weather conditions. However, the information required for these decisions is often distributed across different sources.

One of the major challenges is **crop selection**. Farmers may need to consider various agricultural parameters before selecting a crop. Without an appropriate decision-support mechanism, crop selection may rely primarily on personal experience or generalized information.

Another challenge is obtaining **crop-specific farming guidance**. Farmers may require information related to cultivation practices and crop requirements at different stages of

farming. Accessing relevant information from scattered sources can be inconvenient and time-consuming.

Farm management is also an important requirement. Farmers may need to maintain records of farms, crops, activities, expenses, and production. Manual record keeping can make organizing and reviewing this information difficult.

Furthermore, farmers need access to **market price information** to understand the prevailing prices of agricultural commodities. Market information can be obtained from different sources, but a centralized platform can make access more convenient.

Weather is another important factor that can influence agricultural activities. Farmers require access to current weather conditions and forecasts to support their planning.

Therefore, there is a need for an integrated agricultural platform that can:

- Recommend suitable crops using an AI-based system.
- Provide crop-specific farming guidance.
- Help farmers manage farm-related records and activities.
- Provide agricultural market price information.
- Provide weather conditions and forecast information.
- Present these services through a centralized and user-friendly platform.

KisanSangam aims to address these requirements by combining AI-based crop recommendation with agricultural guidance, farm management, market price analysis, and weather reporting within a single digital platform.

1.3 OBJECTIVES

The main objective of KisanSangam is to develop an intelligent and user-friendly agricultural platform that assists farmers in accessing agricultural information, making informed crop-related decisions, and managing important farming activities.

The key objectives of the project are:

1. To develop a **web-based smart agriculture platform** for providing centralized agricultural services.
2. To develop an **AI-based crop recommendation system** that recommends suitable crops based on relevant agricultural input parameters.

3. To provide a **crop-specific farming guide** containing organized information related to crop cultivation and farming practices.
4. To develop a **farm management module** for maintaining information related to farms, crops, activities, expenses, and production.
5. To provide **market price information** that allows farmers to access relevant agricultural commodity prices.
6. To integrate a **weather reporting system** that provides current weather conditions and forecast information.
7. To integrate the major agricultural functionalities into a **single centralized platform**.
8. To design a **modular and scalable system architecture** that allows the individual components to be developed and maintained efficiently.
9. To provide a **responsive and user-friendly interface** that simplifies interaction with the platform.
10. To demonstrate the practical application of **Artificial Intelligence and modern software technologies in agriculture**.

1.4 SCOPE

1.4 SCOPE

The scope of the KisanSangam project is focused on developing a centralized digital platform that provides agricultural decision-support and farm management functionalities.

The system includes the following major components:

1.4.1 AI-Based Crop Recommendation

The crop recommendation module accepts relevant agricultural parameters as input and processes them using an AI/ML-based model. Based on the provided information, the system generates suitable crop recommendations.

1.4.2 Crop-Specific Farm Guide

The farm guide module provides structured information for different crops. Users can access relevant cultivation information and farming practices associated with the selected crop.

1.4.3 Farm Management

The farm management module enables farmers to maintain and organize important farm-related information. This includes information associated with farms, crops, farming activities, expenses, and production.

1.4.4 Market Price Analysis

The market price module provides information about agricultural commodity prices. It enables farmers to view available market price information and understand prevailing market conditions.

1.4.5 Weather Reporting

The weather reporting module provides weather information for the user's selected location. It can provide current conditions and forecast information that may assist farmers in planning their activities.

Target Users

The primary target users of KisanSangam are **farmers and agricultural users** who require convenient access to agricultural information and farm management facilities.

Platform

The proposed system is developed as a **web-based application** that can be accessed through modern web browsers.

These features may be considered as possible future extensions of the platform

1.5 ORGANIZATION OF REPORT

The KisanSangam project report is organized into multiple chapters to describe the development, implementation, testing, and evaluation of the proposed system.

Chapter 1 – Introduction presents the background of the project, problem statement, objectives, scope, and organization of the report.

Chapter 2 – System Analysis discusses the existing agricultural systems, their limitations, requirements of the proposed system, feasibility analysis, activities of the proposed system, and major system modules.

Chapter 3 – System Design describes the system architecture, design methodology, database design, module design, interface design, and security and access-control mechanisms.

Chapter 4 – Implementation presents the technology stack and implementation of the major KisanSangam modules, including crop recommendation, farm guide, farm management, market price analysis, and weather reporting.

Chapter 5 – Evaluation and Future Work presents the testing and evaluation of the developed system, model performance, system results, limitations, and possible future enhancements.

Chapter 6 – Conclusion and Discussion summarizes the project, discusses the achieved objectives, and presents the overall conclusion of the KisanSangam system.

The report is followed by the **References** and **Appendix**, which contain supporting sources and additional project-related material.

CHAPTER 2: SYSTEM ANALYSIS

2.1 STUDY OF CURRENT SYSTEM

Agricultural information and farming-related services are currently accessed by farmers through a variety of sources, including traditional farming knowledge, agricultural experts, local markets, mobile applications, websites, weather services, and other digital platforms. While these sources provide useful information, the required information is often distributed across different platforms.

Farmers may use separate applications or websites to obtain information about crop cultivation, weather conditions, market prices, and other agricultural activities. Similarly, farm-related records such as crop details, expenses, and farming activities may be maintained manually or through separate record-keeping methods.

One of the important requirements in agriculture is selecting an appropriate crop based on available agricultural conditions. Traditional crop selection may depend on the farmer's experience or advice from other sources. Although experience is valuable, an intelligent recommendation system can assist farmers by analyzing relevant parameters and generating suitable crop recommendations.

Another limitation of existing approaches is that crop-specific farming guidance may not always be available in a structured and easily accessible format. Farmers may need to search through multiple sources to find information related to a particular crop.

Farm management is also frequently performed using manual records. Maintaining information about crops, activities, expenses, and production manually can become difficult as the number of farming activities increases.

Market prices and weather conditions are also important sources of information for farmers. These services may be available through different applications or websites, requiring users to switch between multiple platforms.

The existing situation therefore consists of multiple independent information sources rather than a single integrated platform. KisanSangam addresses this requirement by bringing the major agricultural functionalities into a unified mobile application.

The system consists of an **introductory web page** that presents information about KisanSangam and provides an entry point to the platform, while the main agricultural services are provided through the **KisanSangam mobile application**.

2.2 PROBLEMS AND WEAKNESSES OF CURRENT SYSTEM

The existing approaches to accessing agricultural information have several limitations.

1. Fragmented Information

Agricultural information is often distributed across different applications, websites, local sources, and other platforms. Farmers may need to access multiple sources to obtain the information required for their farming activities.

2. Lack of Personalized Crop Recommendation

Traditional sources may provide general information about crops but may not provide personalized recommendations based on the specific parameters entered by a farmer.

3. Difficult Access to Crop-Specific Guidance

Information regarding cultivation practices may be available through different sources, but finding relevant information for a particular crop can require additional searching and effort.

4. Manual Farm Record Management

Farmers who maintain farm records manually may face difficulties in organizing information related to crops, activities, expenses, and production.

5. Separate Market Information Sources

Market price information may need to be obtained from separate sources. This can make it inconvenient for farmers to access relevant price information alongside their other farming information.

6. Separate Weather Services

Weather information is generally provided through dedicated weather applications or websites. Farmers may need to switch between applications to check weather conditions while using other agricultural services.

7. Limited Integration

The major weakness of the existing approach is the lack of integration between crop recommendation, farming guidance, farm management, market information, and weather reporting.

8. Dependence on Multiple Platforms

Using multiple applications for different agricultural requirements can make the overall process less convenient and less organized for users.

These limitations demonstrate the need for an integrated agricultural application that provides multiple essential functionalities through a single platform.

2.3 REQUIREMENTS OF THE NEW SYSTEM

- Requirement analysis identifies the functional and non-functional requirements that must be satisfied by the proposed KisanSangam system.

2.3.1 Functional Requirements

- The functional requirements of KisanSangam include:
 - **User Management**
 - User registration and login.
 - Secure user authentication.
 - User profile management.
 - Access to personalized farming information.
 - **Crop Recommendation**
 - Accept agricultural parameters from the user.
 - Process the input using the AI/ML model.
 - Generate suitable crop recommendations.
 - Display recommendation results clearly.
 - **Farm Guide**
 - Allow users to select or search for a crop.
 - Display crop-specific farming information.
 - Organize farming guidance in an easily understandable format.
 - **Farm Management**
 - Allow users to create and manage farm records.
 - Add crop information.

- Record farming activities.
- Maintain expense information.
- Maintain production-related information.
- View previously stored farm information.
- **Market Price Analysis**
- Retrieve available agricultural market price information.
- Display crop-related market prices.
- Allow users to view price information for relevant crops or markets.
- **Weather Reporting**
- Allow users to select or provide a location.
- Retrieve weather information for the selected location.
- Display current weather conditions.
- Display available forecast information.
- **Introductory Web Page**
- Present information about KisanSangam.
- Explain the purpose and major capabilities of the platform.
- Provide information about the mobile application.
- Provide an entry point for users interested in the application.

2.3.2 Non-Functional Requirements

- The non-functional requirements of KisanSangam include:
- **Usability:** The application should have a simple and intuitive interface suitable for users with varying levels of technical knowledge.
- **Performance:** The application should provide responses within a reasonable time under normal operating conditions.
- **Reliability:** The system should operate consistently and minimize failures during normal usage.
- **Security:** User accounts and stored farm information should be protected from unauthorized access.
- **Scalability:** The system architecture should support future expansion of users and features.
- **Maintainability:** The application should use a modular structure so individual components can be maintained independently.

- **Responsiveness:** The introductory website should work properly across commonly used desktop and mobile browsers.
- **Availability:** The mobile application should provide access to the core agricultural services whenever the required network services are available.

2.4 SYSTEM FEASIBILITY

Feasibility analysis determines whether the proposed KisanSangam system is practical and achievable from technical, economic, and operational perspectives.

2.4.1 Technical Feasibility

KisanSangam is technically feasible because the required functionality can be developed using currently available mobile development technologies, web technologies, databases, AI/ML frameworks, APIs, and cloud services.

The AI-based crop recommendation component can be developed using machine learning techniques and relevant agricultural datasets. The market price and weather components can obtain information through suitable APIs or external data services.

The mobile application can communicate with backend services through APIs, while the introductory web page can provide information about the platform independently.

Therefore, the required technologies are available and the proposed system is technically feasible.

2.4.2 Economic Feasibility

The proposed system is economically feasible because the development can make use of open-source programming languages, frameworks, libraries, and development tools.

The primary development costs may involve cloud hosting, API services, database services, deployment, and other infrastructure requirements. These costs can be controlled during the development and academic project phase by using suitable free or low-cost service tiers where applicable.

Therefore, the project can be developed within a reasonable academic project budget.

2.4.3 Operational Feasibility

KisanSangam is designed to provide a simple mobile interface through which users can access agricultural services.

The major features are organized into separate modules, allowing users to access crop recommendations, farming guidance, farm management, market prices, and weather information without requiring advanced technical knowledge.

The introductory website provides users with an overview of the platform before they use the mobile application.

Therefore, the proposed system is operationally feasible.

2.5 ACTIVITIES IN PROPOSED SYSTEM

The proposed KisanSangam system performs a series of activities to provide agricultural services to users.

Initially, users can visit the **KisanSangam introductory web page** to understand the purpose and major functionalities of the platform. The web page acts as the public-facing introduction to the application.

Users can then access the KisanSangam mobile application and complete the required registration and authentication process.

After successful authentication, the user is provided with access to the application's main dashboard.

From the dashboard, users can access the major modules of the system.

Crop Recommendation Activity

The user provides the required agricultural parameters to the crop recommendation module. The system processes these parameters through the trained AI/ML model and generates suitable crop recommendations.

Farm Guide Activity

The user selects a crop for which farming information is required. The system retrieves the corresponding crop-specific information and presents it in an organized format.

Farm Management Activity

The user can create and maintain farm-related records. Information such as farm details, crops, activities, expenses, and production can be entered and stored for future reference.

Market Price Activity

The user selects a crop or relevant market information and accesses the available market price data. The system displays the retrieved information in an understandable format.

Weather Reporting Activity

The user provides or selects a location. The system retrieves the relevant weather information and displays current conditions and available forecast information.

These activities work together to provide an integrated agricultural platform rather than requiring the user to depend on multiple independent applications.

2.6 MAIN MODULES

The KisanSangam system is divided into several modules. Each module performs a specific function within the platform.

1. User Authentication Module

The authentication module manages user registration and login. It verifies user credentials and provides authenticated access to the mobile application.

2. Crop Recommendation Module

This is the primary AI-based module of the system. It accepts relevant agricultural parameters and processes them using a machine learning model to generate suitable crop recommendations.

3. Farm Guide Module

The farm guide module provides crop-specific agricultural information. Users can select a crop and access relevant cultivation guidance in an organized format.

4. Farm Management Module

The farm management module allows users to maintain digital records associated with their farms. It can manage farm details, crop records, activities, expenses, and production information.

5. Market Price Module

The market price module provides agricultural commodity price information. It retrieves and displays available market-related data to the user.

6. Weather Reporting Module

The weather reporting module provides weather information based on the user's selected location. It displays current conditions and available forecast information.

7. Mobile Application Module

The mobile application acts as the primary user interface through which farmers interact with the core KisanSangam services.

8. Introductory Web Module

The introductory web module provides a public-facing web page that introduces KisanSangam, explains its purpose, and presents its major capabilities.

Table 2.1: System Module Descriptions

Module	Description
User Authentication	Manages user registration, login, and authentication.
Crop Recommendation	Uses an AI/ML model to recommend suitable crops based on user-provided parameters.
Farm Guide	Provides crop-specific farming and cultivation information.
Farm Management	Maintains farm, crop, activity, expense, and production records.
Market Price	Provides agricultural commodity market price information.
Weather Reporting	Displays current and forecast-based weather information.
Mobile Application	Provides the primary interface for accessing KisanSangam services.
Introductory Web Page	Introduces the KisanSangam platform and its major capabilities.

The modular structure follows the reference report's approach of dividing the application into separate functional modules with clearly defined responsibilities.

MindClash report chapters

2.7 SOFTWARE AND HARDWARE

2.7.1 Software Requirements

The exact technology stack can be finalized according to the implementation, but the KisanSangam system requires the following categories of software:

- **Operating System:** Windows, Linux, macOS, Android
- **Mobile Application Development:** Mobile application framework
- **Web Technologies:** HTML, CSS, JavaScript/TypeScript
- **Backend:** Server-side framework for API development
- **Database:** Database management system for storing application and farm data
- **Machine Learning:** Python, NumPy, Pandas, Scikit-learn and related ML libraries
- **AI/ML Model:** Trained crop recommendation model
- **APIs:** REST APIs for communication between application components
- **Weather Service:** Weather API
- **Market Data Service:** Appropriate market/price data source or API
- **Development Environment:** Visual Studio Code or equivalent IDE
- **Version Control:** Git and GitHub
- **Testing Tools:** API and application testing tools
- **Web Browser:** Google Chrome, Microsoft Edge, Mozilla Firefox

2.7.2 Hardware Requirements

Development System

- **Processor:** Intel Core i5 / AMD equivalent or higher
- **RAM:** Minimum 8 GB recommended
- **Storage:** Minimum 10 GB available for project development
- **Internet Connection:** Required for APIs, cloud services, development resources, and deployment
- **Display:** Standard HD display

End-User Device

The primary end-user platform is a **smartphone capable of running the KisanSangam mobile application.**

A stable internet connection may be required for services such as market price retrieval, weather reporting, authentication, and server-based operations.

2.8 PROPOSED SYSTEM OVERVIEW

The proposed KisanSangam system consists of two primary user-facing components:

1.Introductory Web Page

The web page acts as the public introduction to KisanSangam. It communicates the purpose of the platform and provides information about its major features.

2.Mobile Application

The mobile application contains the core agricultural functionalities used by farmers.

The overall interaction can be represented as:

**User → Introductory Web Page → KisanSangam Mobile App → Backend APIs
→ AI/ML Model / Database / External Services**

The mobile application communicates with backend services to process user requests, store farm information, execute the crop recommendation model, and retrieve external information such as market prices and weather data.

This structure provides a clear separation between the public introductory interface and the core application functionality.

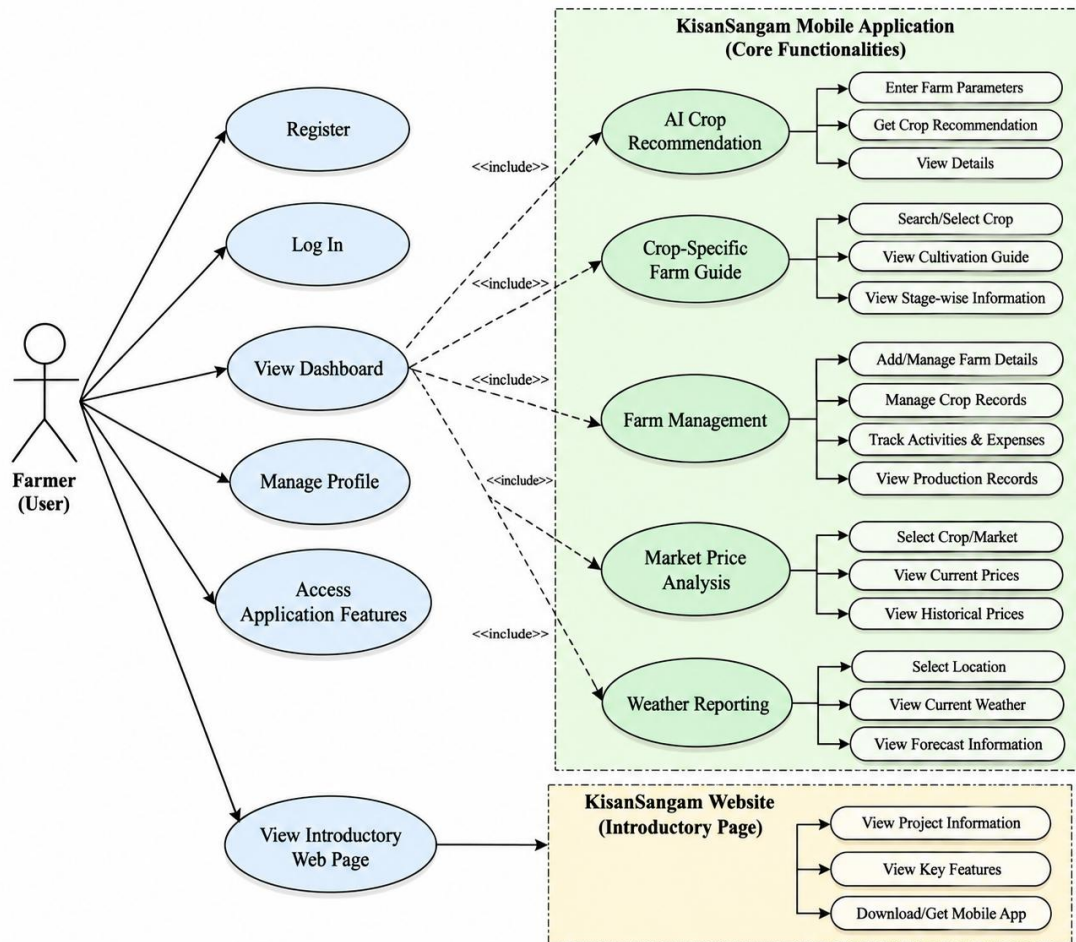


Figure 2.1: Use Case Diagram for MindClash

CHAPTER 3: SYSTEM DESIGN

3.1 DESIGN METHODOLOGY

The development of the KisanSangam system follows an iterative and modular development approach. The project is divided into independent modules so that each major functionality can be developed, tested, and integrated systematically.

The core functionalities of **KisanSangam** include AI-Based Crop Recommendation, Crop-Specific Farm Guide, Farm Management, Market Price Analysis, and Weather Reporting. Each module is designed to perform a specific function while communicating with the backend services and database where required.

The system is designed around a mobile-first architecture because the KisanSangam mobile application is the primary platform for farmers. An introductory web page is provided separately to present information about the project and its major features.

The overall system can be organized into three major logical layers:

- **Presentation Layer**
- **Application Layer**
- **Data Layer**

The reference report similarly separates its system into presentation, application, and data tiers to improve maintainability and scalability.

Presentation Layer

The presentation layer consists of the KisanSangam mobile application through which farmers interact with the system. It provides interfaces for authentication, crop recommendation, farm guidance, farm management, market prices, and weather reporting.

The introductory web page acts as the public-facing interface of the project and provides information about KisanSangam and its major functionalities.

Application Layer

The application layer contains the backend services and business logic of the system. It processes requests received from the mobile application, manages authentication,

communicates with the database, processes farm-related operations, and connects with the AI/ML model and external data services.

Data Layer

The data layer is responsible for storing and retrieving application data. It stores information such as user profiles, farm details, crop records, farming activities, expenses, and other persistent information required by the application.

The architecture allows each layer to be developed and maintained independently, making the system easier to modify and expand.

3.2 SYSTEM ARCHITECTURE

The KisanSangam system follows a layered architecture in which the mobile application communicates with backend services through APIs. The backend processes user requests and communicates with the appropriate database, AI/ML model, or external service.

The major components of the proposed architecture are:

1. KisanSangam Mobile Application

The mobile application is the primary interface for farmers. It provides access to all five core functionalities:

- AI-Based Crop Recommendation
- Crop-Specific Farm Guide
- Farm Management
- Market Price Analysis
- Weather Reporting

2. Introductory Web Page

The introductory web page provides information about the KisanSangam platform, its objectives, and its major features. It acts as the initial public-facing interface for the project.

3. Backend/API Layer

The backend provides APIs through which the mobile application communicates with the server. It handles authentication, business logic, data processing, and communication with external services.

4. AI/ML Service

The AI/ML component processes agricultural input parameters provided by the farmer and generates crop recommendations using the trained crop recommendation model.

5. Database

The database stores persistent information required by the system, including user details, farm records, crop information, activities, expenses, and production records.

6. External Data Services

External services are used where required for functionalities such as:

- Weather reporting
- Market price information

The overall communication flow can be represented as:

Farmer → Mobile Application → REST API → Backend → Database / AI Model / External APIs → Backend → Mobile Application

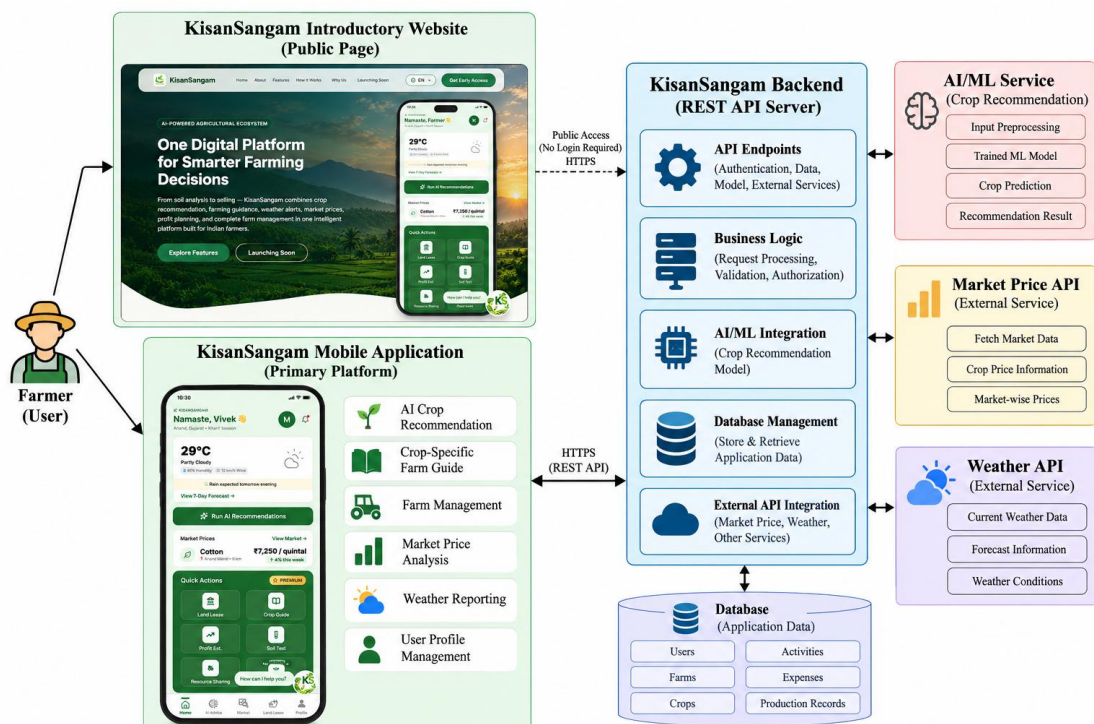


Figure 3.1: KisanSangam System Architecture

3.3 SYSTEM MODULE DESIGN

The KisanSangam system is divided into functional modules. Each module is responsible for a particular system operation.

Table 3.1: System Module Descriptions

Module	Description
User Authentication	Manages registration, login, and secure access to the application.
AI Crop Recommendation	Processes agricultural parameters and recommends suitable crops using an ML model.
Crop-Specific Farm Guide	Provides structured farming and cultivation information for selected crops.
Farm Management	Allows farmers to manage farm, crop, activity, expense, and production records.
Market Price Analysis	Provides relevant agricultural commodity market price information.
Weather Reporting	Provides current weather conditions and forecast information for a selected location.
Database Management	Stores and retrieves user and farm-related information.
API Management	Handles communication between the mobile application, backend, database, AI model, and external services.

The modular organization follows the reference report's approach of assigning a clear responsibility to each system module. MindClash report chapters

3.4 AI CROP RECOMMENDATION DESIGN

The AI-based crop recommendation system is one of the major components of KisanSangam.

The module receives relevant agricultural parameters from the farmer. These parameters are processed and passed to the trained machine learning model.

The general workflow is:

User Input → Input Validation → Data Preprocessing → ML Model → Crop Prediction → Recommendation Display

The system first collects the required input values from the user. The input is then validated to ensure that the supplied values are suitable for processing.

After preprocessing, the data is passed to the trained crop recommendation model. The model analyzes the input features and generates a predicted crop recommendation.

The prediction is then returned to the mobile application and displayed to the farmer in an understandable format.

AI Crop Recommendation Workflow

1. Farmer opens the Crop Recommendation module.
2. Farmer enters the required agricultural parameters.
3. System validates the input.
4. Input data is preprocessed according to the model requirements.
5. Processed data is sent to the ML model.
6. The model generates the crop prediction.
7. The prediction is returned to the application.
8. Recommended crop information is displayed to the farmer.

The specific model, dataset, input parameters, preprocessing technique, and evaluation metrics will be documented in **Chapter 4: Implementation**, based on the actual model used in KisanSangam.

3.5 CROP-SPECIFIC FARM GUIDE DESIGN

The Crop-Specific Farm Guide provides organized agricultural information for selected crops.

The module is designed to allow users to select or search for a crop and then access relevant farming information.

The general workflow is:

Select Crop → Retrieve Crop Information → Organize Information → Display Farming Guide

The guide may organize information into relevant categories such as:

- Crop overview
- Cultivation requirements
- Sowing information
- Growth stages
- Irrigation requirements
- Nutrient requirements
- General cultivation practices
- Harvesting information

The exact content displayed by the module will depend on the agricultural information incorporated into the KisanSangam system.

The purpose of the module is to provide farmers with a structured source of crop-specific information rather than requiring them to search across multiple sources.

3.6 FARM MANAGEMENT DESIGN

The Farm Management module is designed to help farmers digitally maintain information related to their farming activities.

The module can maintain records associated with:

- Farm details
- Crop details
- Farming activities
- Expenses
- Production records

The general workflow is:

Add Farm → Add Crop → Record Activities → Record Expenses → Record Production → View/Manage Records

Farmers can create farm records and associate crop information with the corresponding farm. Activities and expenses can then be recorded and reviewed through the application.

The module provides a structured approach to maintaining farm information and reduces dependence on manually maintained records.

Farm Management Operations

The module supports operations such as:

- Adding a farm
- Updating farm information
- Adding crop records
- Updating crop records
- Recording farming activities
- Recording expenses

- Recording production information
- Viewing stored records

The database relationships for these records will be represented through the **Entity-Relationship Diagram** in the following section.

3.7 MARKET PRICE ANALYSIS DESIGN

The Market Price Analysis module provides agricultural commodity price information to users.

The module communicates with the relevant market-price data source through an API or appropriate data service.

The general workflow is:

Select Crop/Market → API Request → Retrieve Price Data → Process Data → Display Market Information

The system retrieves the available price information and presents it through the mobile application.

The module is intended to provide farmers with convenient access to market-related information without requiring them to use a separate application for checking crop prices.

Depending on the available data source, the system can display relevant information such as:

- Crop/commodity name
- Market
- Price
- Date
- Other available market information

The exact data fields will depend on the market data source used during implementation.

3.8 WEATHER REPORTING DESIGN

The Weather Reporting module provides weather information based on the location selected or provided by the user.

The general workflow is:

Select Location → Weather API Request → Retrieve Weather Data → Process Response → Display Weather Report

The system communicates with the weather service through an API and retrieves the available weather information.

The mobile application can present information such as:

- Current temperature
- Weather condition
- Humidity
- Wind information
- Rainfall/precipitation information
- Available forecast information

The weather module is designed as a **reporting feature** rather than an automated agricultural alert or recommendation system.

3.9 DATABASE DESIGN

The database is responsible for storing persistent information required by the KisanSangam application.

The major entities can include:

- **Users**
- **Farms**
- **Crops**
- **Farm Activities**
- **Expenses**
- **Production Records**

The database structure is designed to maintain logical relationships between these entities.

For example:

- A user can have one or more farms.
- A farm can contain multiple crop records.
- A crop record can have multiple farming activities.
- A farm can have multiple expense records.
- A farm or crop can have associated production records.

Table 3.2: Major Database Entities

Entity	Purpose
Users	Stores user account and profile information.
Farms	Stores information about farms managed by the user.
Crops	Stores crop-related records associated with farms.
Activities	Stores farming activities performed for crops/farms.
Expenses	Stores farm-related expenditure records.
Production	Stores production-related information.

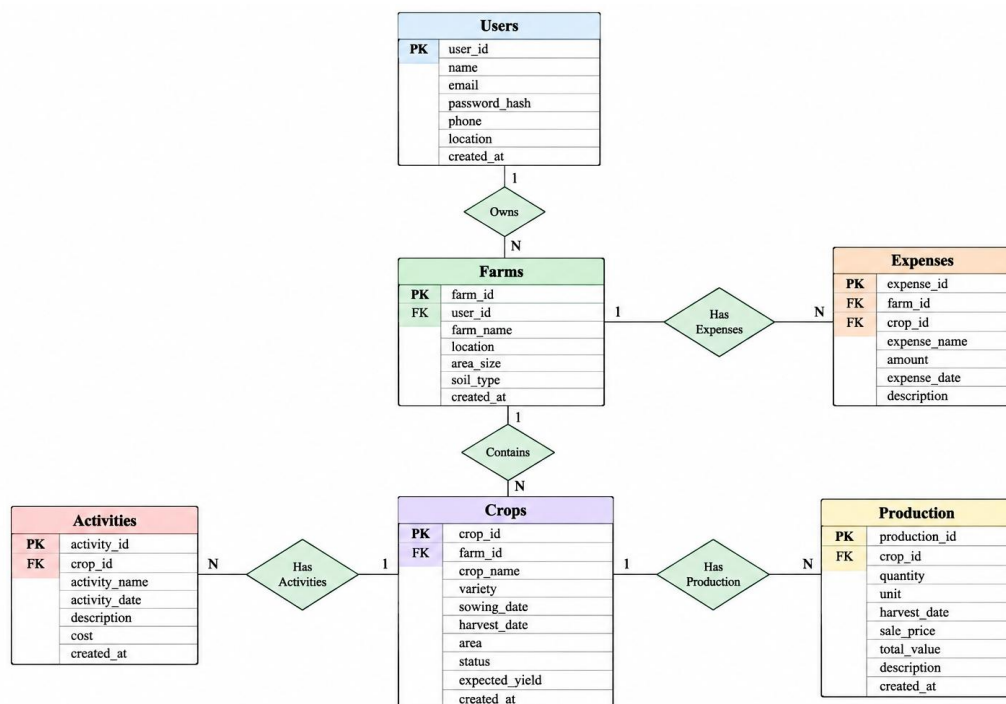


Figure 3.2: Entity-Relationship Diagram

The ER diagram should illustrate the relationships between:

User → Farm → Crop → Activities / Expenses / Production

The reference report similarly uses an ER diagram to show relationships between its principal database entities. MindClash report chapters

3.10 INTERFACE DESIGN

The KisanSangam user interface is designed with a focus on simplicity, accessibility, and ease of navigation.

Since the **mobile application is the primary platform**, the interface is designed around mobile screens and touch-based interaction.

The major interfaces include:

3.10.1 Introductory Web Page

The introductory web page provides:

- KisanSangam introduction
- Project overview
- Key features
- Information about the mobile application
- Application access/download information

3.10.2 Registration and Login Interface

The authentication interface allows users to create accounts and securely access the application.

3.10.3 Dashboard

The dashboard acts as the central navigation interface for the mobile application.

It provides access to:

- Crop Recommendation
- Farm Guide
- Farm Management
- Market Prices
- Weather Reporting

3.10.4 Crop Recommendation Interface

This interface allows farmers to enter the parameters required by the crop recommendation model and view the resulting recommendation.

3.10.5 Farm Guide Interface

Users can search for or select a crop and view the corresponding farming information.

3.10.6 Farm Management Interface

Users can add and manage their farm information, crops, activities, expenses, and production records.

3.10.7 Market Price Interface

Users can select a crop or market and view available market price information.

3.10.8 Weather Interface

Users can select a location and view the available weather information and forecast.

The reference report similarly separates registration, dashboard, and feature-specific interfaces when describing its interface design. MindClash report chapters

3.11 SECURITY AND ACCESS CONTROL

Security is an important consideration in KisanSangam because the application stores user account information and farm-related records.

The system can incorporate the following security mechanisms:

1. User Authentication

Users must authenticate themselves before accessing protected application functionality.

2. Secure Password Management

User passwords should not be stored in plain text. Password hashing mechanisms should be used to protect authentication credentials.

3. Authorization

The backend should verify that authenticated users can access and modify only the records associated with their accounts.

4. API Security

Backend APIs should validate authentication credentials and user requests before processing protected operations.

5. Input Validation

Inputs received from the mobile application should be validated before being processed or stored in the database.

6. Data Protection

Farm records and user information should be protected from unauthorized access and modification.

7. Secure Communication

Communication between the mobile application and backend services should use secure network protocols such as HTTPS.

These mechanisms follow the general security principles described in the reference report, including authentication, access control, input validation, and secure deployment. MindClash report chapters

3.12 SUMMARY OF SYSTEM DESIGN

The KisanSangam system is designed as a modular agricultural platform consisting of a **mobile application as the primary user interface** and an **introductory web page** as the public-facing project interface.

The system uses a layered architecture consisting of presentation, application, and data components. The mobile application communicates with backend APIs, while the backend manages the AI/ML model, database operations, and external services.

The five primary functional modules—**AI Crop Recommendation, Crop-Specific Farm Guide, Farm Management, Market Price Analysis, and Weather Reporting**—are designed to work together while maintaining independent responsibilities.

This modular architecture provides a foundation for maintainability, scalability, and future expansion of the KisanSangam platform.

CHAPTER 4: IMPLEMENTATION

4.1 TECHNOLOGY STACK

The KisanSangam platform is developed using a combination of modern technologies for mobile application development, backend services, artificial intelligence, database management, API integration, and user-interface design. The technology stack is divided into different components according to their role in the system.

4.1.1 Mobile Application Development

The KisanSangam farmer application is developed for mobile devices using Kotlin. Kotlin provides a modern programming environment for Android application development and is used to implement the application's screens, navigation, user interactions, and communication with backend services.

4.1.2 Backend Development

The backend provides the central application services and REST APIs required by the mobile application. It handles authentication, business logic, data processing, database communication, and integration with external services.

Major technologies used for backend development include:

- Python
- FastAPI
- REST APIs
- Unicorn

Python is also used for the AI/ML components of the project.

4.1.3 Artificial Intelligence and Machine Learning

Python-based machine learning technologies are used for the AI Crop Recommendation functionality.

The major libraries and tools include:

- Python
- NumPy
- Pandas

- Scikit-learn
- Matplotlib
- Seaborn
- Plotly

These technologies support data preprocessing, analysis, visualization, model development, and evaluation.

4.1.4 Database

A database is used to store user and farm-related information required by the application. The database maintains information such as users, farms, crops, activities, expenses, and production records.

4.1.5 UI/UX Design

Figma is used for designing the user interface and user experience of the KisanSangam platform.

The UI/UX design process includes:

- Wireframing
- Screen layout design
- Navigation planning
- Mobile interface design
- Component design
- Prototype development
- User-flow design

The final mobile interface is implemented based on the designed UI/UX structure.

4.1.6 Web Technologies

The introductory KisanSangam website uses standard web technologies for creating the public-facing interface.

The major technologies include:

- HTML
- CSS
- JavaScript

These technologies are used to implement the introductory website, navigation, feature presentation, and responsive interface elements.

4.1.7 API Integration

External APIs are integrated into the platform to provide dynamic agricultural information.

The major integrations include:

- Weather API – weather and forecast information
- Market Price API/Service – crop market price information
- REST APIs – communication between the mobile application and backend

4.1.8 Development and Deployment Tools

The development process uses modern software-development tools for coding, testing, version control, and deployment.

These include:

- Visual Studio Code
- Android Studio
- Git
- GitHub
- Postman
- Docker
- Unicorn

Table 4.1: Technology Stack and Tools

Component	Technology / Tool	Purpose
Mobile Application	Kotlin	Android application development
Android Development	Android Studio	Mobile application development and testing
UI/UX Design	Figma	Interface, wireframes, prototypes and user flows
Web	HTML, CSS, JavaScript	Introductory website
Backend Language	Python	Backend and AI/ML development
Backend Framework	FastAPI	REST API development

Component	Technology / Tool	Purpose
Application Server	Uvicorn	Running the FastAPI application
Machine Learning	Scikit-learn	Crop recommendation model
Data Processing	Pandas, NumPy	Data processing and preprocessing
Data Visualization	Matplotlib, Seaborn, Plotly	Data analysis and visualization
Database	Project Database	User and farm data storage
Weather Service	Weather API	Weather and forecast information
Market Service	Market Price API/Service	Market price information
API Testing	Postman	API testing
Version Control	Git, GitHub	Source-code management
Containerization	Docker	Application packaging and deployment
Code Editor	Visual Studio Code	Software development

4.2 MODULE DEVELOPMENT

The KisanSangam system is divided into multiple modules, with each module responsible for a specific functionality of the platform. The modules are integrated through the mobile application, backend services, database, AI/ML component, and external APIs.

4.2.1 User Authentication Module

The User Authentication Module manages the registration and login process of farmers. It verifies the credentials provided by the user and provides authenticated access to the application.

The module handles:

- User registration
- User login
- Credential validation
- User profile information
- Authentication and authorized access

After successful authentication, the user is directed to the KisanSangam mobile dashboard.

4.2.2 AI Crop Recommendation Module

The AI Crop Recommendation Module uses a trained machine learning model to provide crop recommendations based on agricultural parameters provided by the farmer.

The implementation follows the process:

Input Parameters → Data Preprocessing → ML Model → Prediction → Crop Recommendation

The required input is collected through the mobile application and sent to the backend. The backend processes the request and communicates with the ML model. The generated recommendation is then returned to the application and displayed to the farmer.

4.2.3 Crop-Specific Farm Guide Module

The Crop-Specific Farm Guide module provides agricultural information related to the selected crop.

The farmer can select a crop through the application and access the corresponding crop information. The module retrieves the relevant data and presents it through the mobile interface in an organized format.

The general workflow is:

Select Crop → Retrieve Crop Information → Process Data → Display Information

4.2.4 Farm Management Module

The Farm Management Module allows farmers to digitally maintain information related to their farms and crops.

The module supports management of:

- Farm details
- Crop records
- Farming activities
- Expenses
- Production records

The information entered by the farmer is sent to the backend and stored in the database. Stored records can subsequently be retrieved and displayed through the application.

The general workflow is:

Add Farm → Add Crop → Record Activities → Record Expenses → Record Production → View Records

4.2.5 Market Price Analysis Module

The Market Price Analysis Module provides market price information for agricultural crops.

The farmer selects the required crop or market information through the application. The backend communicates with the integrated market-price service, processes the received information, and returns the relevant data to the mobile application.

The workflow is:

Select Crop/Market → API Request → Retrieve Market Data → Process Response → Display Price

This module allows market-related information to be accessed within the KisanSangam platform.

4.2.6 Weather Reporting Module

The Weather Reporting Module provides current weather conditions and forecast information.

The application sends the required location information to the backend. The backend communicates with the integrated weather service and processes the received weather data before returning it to the application.

The workflow is:

Location → Weather API Request → Retrieve Weather Data → Process Response → Display Weather Information

The current implementation treats this functionality as weather reporting, providing weather information and forecast data to the user.

4.2.7 Introductory Website Module

The Introductory Website Module represents the public-facing web interface of KisanSangam.

The website provides information about:

- KisanSangam
- Major platform features
- How the platform works
- Benefits of the platform
- Application access information

The website acts as the introductory point for users before accessing the primary mobile application.

4.2.8 Mobile Application Dashboard Module

The Mobile Application Dashboard acts as the central interface of the KisanSangam application.

The dashboard provides access to important information and the major application modules. It brings together relevant information such as weather and market information while providing navigation to the agricultural functionalities.

The dashboard is designed using the UI/UX structure developed in Figma and implemented in the mobile application using Kotlin.

Table 4.2: Module Description

Module	Description
User Authentication Module	Manages registration, login, authentication, and user access.
AI Crop Recommendation Module	Processes agricultural parameters and generates crop recommendations using the ML model.
Crop-Specific Farm Guide Module	Provides information related to selected crops.
Farm Management Module	Manages farm, crop, activity, expense, and production records.
Market Price Analysis Module	Retrieves and displays agricultural market price information.
Weather Reporting Module	Retrieves and displays current weather and forecast information.
Introductory Website Module	Provides public information about the KisanSangam platform.
Mobile Dashboard Module	Provides the central interface for accessing application functionality.

4.3 TESTING

Testing was carried out to verify the correctness, reliability, and integration of the KisanSangam system. The testing process focused on the major modules of the application, backend API communication, AI/ML functionality, database operations, and external service integration.

The testing was performed at different levels to identify functional and integration issues during development.

4.3.1 Unit Testing

Unit testing was performed on individual functions and components of the system to verify that they produced the expected results independently.

The testing included:

- User input validation
- Authentication functions
- Crop recommendation processing
- Farm record operations
- Expense and production record operations

- Data processing functions
- Weather and market data processing

Each individual component was tested with valid and invalid inputs wherever applicable.

4.3.2 Integration and API Testing

Integration testing was performed to verify communication between the mobile application, backend, database, AI/ML model, and external services.

The API testing process verified:

- Correct request and response handling
- Authentication requests
- Farm and crop data operations
- AI crop recommendation requests
- Market price API communication
- Weather API communication
- Database operations
- Error handling for invalid requests

Postman was used for testing backend API requests and responses during development.

4.3.3 Functional Testing

Functional testing was performed by executing the main workflows of the KisanSangam application.

The following operations were tested:

- User registration
- User login
- Dashboard access
- AI crop recommendation
- Crop guide access
- Adding farm information
- Adding crop records
- Recording farming activities
- Recording expenses

- Recording production information
- Viewing market prices
- Viewing weather information

The objective was to verify that each feature performs according to its intended functionality.

4.3.4 User Interface Testing

The user interface was tested to verify the usability and consistency of the introductory website and mobile application.

The following aspects were checked:

- Screen navigation
- Button functionality
- Input fields
- Screen layout
- Information visibility
- Navigation between modules
- Mobile responsiveness
- Consistency of UI components

The interface was designed using Figma and implemented in the mobile application using Kotlin.

4.3.5 Performance Testing

Performance testing was performed to observe the application's behavior during normal operations.

The testing focused on:

- API response handling
- Database request processing
- AI model response time
- External API communication
- Mobile application responsiveness
- Loading and navigation between screens

The modular architecture allows individual components to be optimized independently when required.

4.3.6 Security Testing

Basic security testing was performed to verify that protected application functions are accessible only to authorized users.

The testing included:

- Authentication validation
- Invalid credential handling
- Input validation
- Unauthorized-access checks
- API request validation
- Protection of user-related information

These checks help reduce the possibility of invalid or unauthorized operations within the system.

Table 4.3: Sample Test Results

Test Case	Expected Result	Actual Result	Status
User registration with valid data	Account should be created	Account created successfully	Pass
Login with valid credentials	User should access dashboard	Dashboard displayed	Pass
Login with invalid credentials	Error should be displayed	Error displayed	Pass
AI crop recommendation	Crop recommendation should be generated	Recommendation displayed	Pass
Open crop guide	Crop information should be displayed	Crop information displayed	Pass
Add farm details	Farm record should be stored	Farm record stored	Pass
Add crop record	Crop record should be stored	Crop record stored	Pass
Record expense	Expense should be stored	Expense recorded	Pass
View market price	Market information should be displayed	Market information displayed	Pass
View weather	Weather information should be displayed	Weather information displayed	Pass

The testing process helped verify the major functionalities and interactions within the KisanSangam system. Identified issues during development were addressed through debugging, API testing, and repeated functional testing.

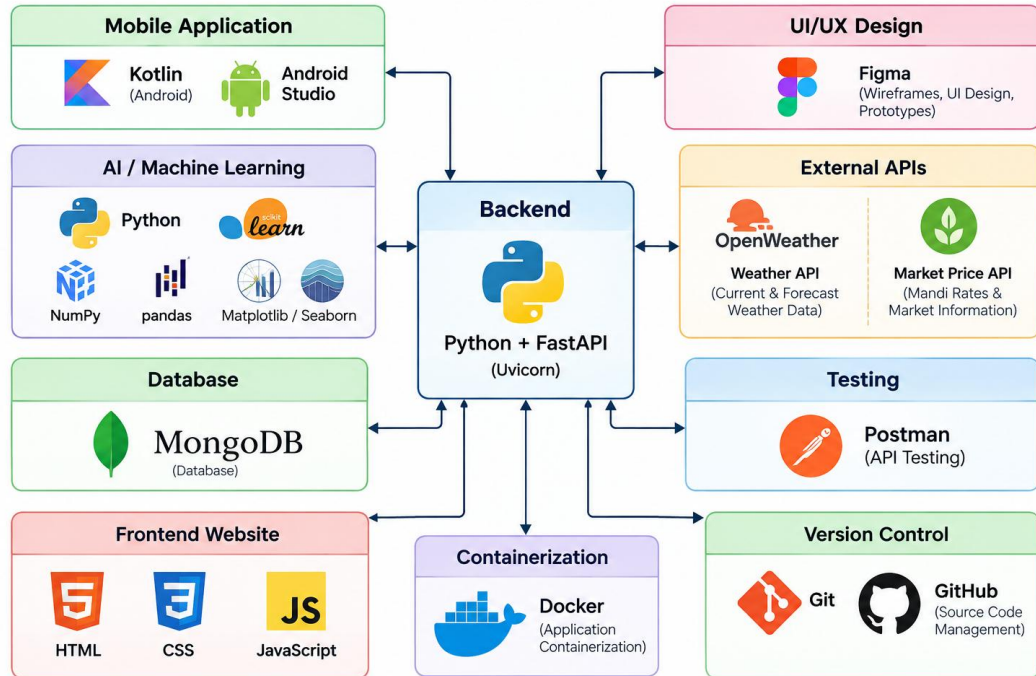


Figure 4.1: Technology Stack and Tools

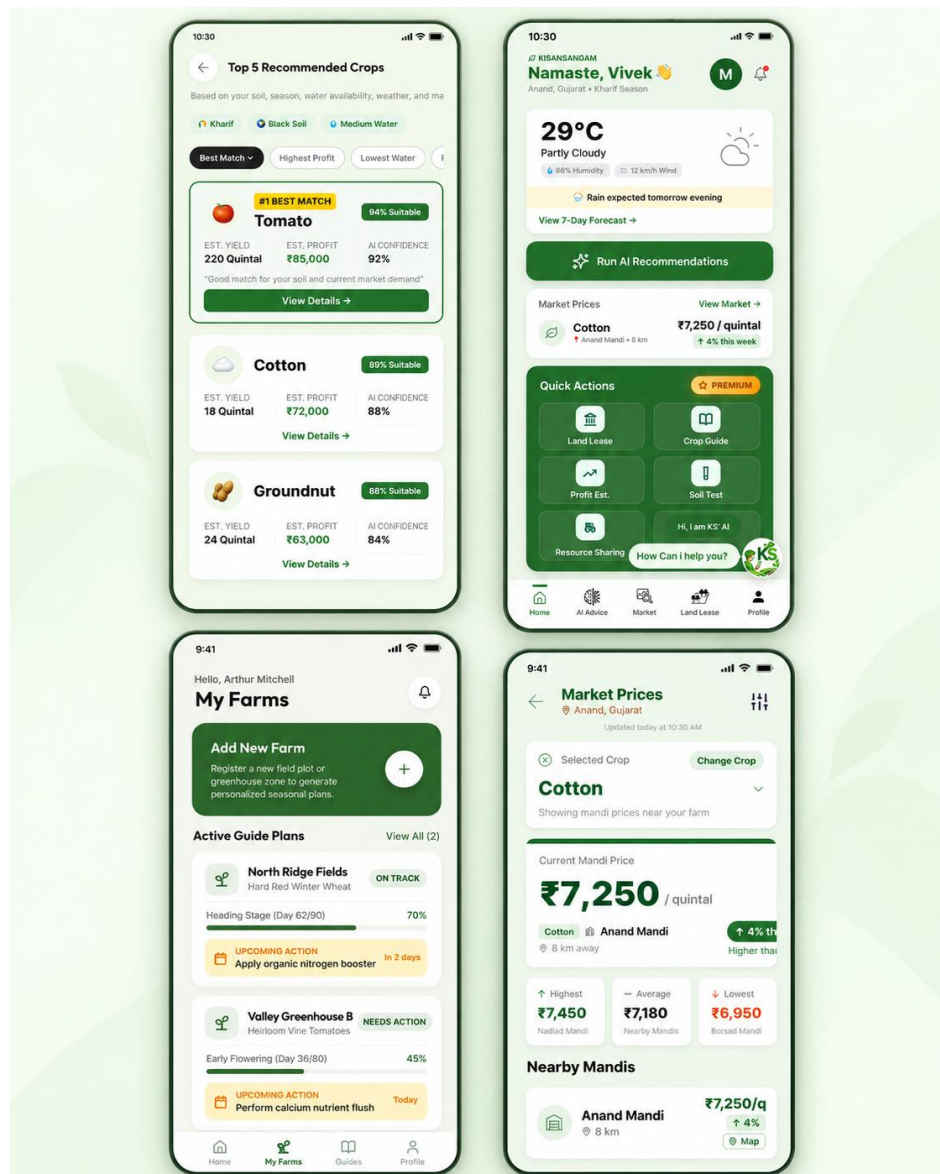


Figure 4.2: Home Page Implementation Image

CHAPTER 5: EVALUATION AND FUTURE WORK

5.1 EVALUATION OF RESULTS

The KisanSangam system was evaluated based on its functionality, usability, performance, and reliability. The objective of the evaluation was to verify whether the implemented system satisfies the requirements defined during the system analysis and design stages.

The evaluation focused on the five core functionalities of the platform: AI Crop Recommendation, Crop-Specific Farm Guide, Farm Management, Market Price Analysis, and Weather Reporting.

Functional Evaluation

The functional evaluation was performed to verify the operation of the major modules of KisanSangam.

The AI Crop Recommendation module processes the agricultural parameters provided by the user and generates a crop recommendation through the implemented machine learning model.

The Crop-Specific Farm Guide module provides relevant information for the crop selected by the user.

The Farm Management module allows users to maintain farm-related records, including farm details, crop records, activities, expenses, and production information.

The Market Price Analysis module retrieves and displays relevant crop market price information through the integrated market data service.

The Weather Reporting module retrieves and displays current weather conditions and forecast information through the integrated weather service.

The introductory website and mobile application were also evaluated to verify navigation and accessibility of the implemented features.

Overall, the functional evaluation verifies the integration of the major modules into a single agricultural platform.

Performance Evaluation

The performance evaluation focused on the communication between the mobile application, backend services, database, AI/ML component, and external APIs.

The modular architecture separates user-interface operations from backend processing and data management. This allows different components of the system to handle their respective tasks independently.

The AI crop recommendation process involves input processing followed by model prediction, while market price and weather information are obtained through their respective external services.

The system architecture also provides scope for further performance optimization through improvements in API communication, database operations, model processing, and server-side resource management.

Usability Evaluation

The usability evaluation focused on the ease of interaction with the KisanSangam platform.

The mobile application provides a centralized dashboard through which users can access the major agricultural functionalities. The introductory website provides information about KisanSangam and its major features.

The application is designed with a straightforward navigation structure so that users can access agricultural information and manage their farm-related records from a single platform.

Reliability Evaluation

The reliability evaluation focused on the consistent operation of the major system components.

The backend acts as the central communication layer between the mobile application, database, AI/ML component, and external services. The database provides persistent storage for user and farm-related information.

Authentication and input validation mechanisms are incorporated to ensure that application requests are processed appropriately.

The modular architecture also allows individual components to be modified or maintained without requiring significant changes to the complete system.

Table 5.1: Evaluation of Major Modules

Module	Evaluation Criteria	Expected Outcome
AI-Crop Recommendation	Agricultural input processing and prediction	Crop recommendation is generated
Crop-Specific Farm Guide	Crop information retrieval	Relevant crop information is displayed
Farm Management	Record creation and retrieval	Farm records are stored and managed
Market Price Analysis	Market data integration	Market price information is displayed
Weather Reporting	Weather data integration	Weather information is displayed
Mobile Application	Navigation and usability	Users can access major functions
Introductory Website	Information presentation	Platform information is displayed
Authentication	Credential verification	Authorized users can access protected features

5.2 FUTURE ENHANCEMENTS

Although the current version of KisanSangam focuses on its five core functionalities, the platform can be further enhanced by integrating additional technologies and capabilities.

5.2.1 Soil Testing Integration

Future versions of KisanSangam can integrate soil-testing capabilities with the platform. Data obtained from compatible soil-testing devices can be incorporated into the system and used as additional agricultural information for analysis and crop recommendation.

5.2.2 Advanced AI and Data Analytics

The AI capabilities of KisanSangam can be further improved by using larger agricultural datasets and additional data sources. Future development can focus on improving the accuracy and reliability of AI-based agricultural recommendations and generating more meaningful insights from collected farm data.

5.2.3 Multilingual Support

The platform can be extended to support multiple Indian languages. Providing regional language support can make the application more accessible to farmers from different regions.

Languages such as Gujarati and Hindi can be considered for future implementation.

5.2.4 Cloud Scalability

Future versions can be deployed using scalable cloud infrastructure to support a larger number of users and increasing amounts of agricultural data.

Cloud-based deployment can provide improved scalability, availability, storage capacity, and system management.

5.2.5 Subscription and Service Expansion

A future version of KisanSangam can introduce subscription-based services for selected advanced functionalities. This can provide a potential sustainable model for maintaining and expanding the platform.

5.2.6 Additional Agricultural Data Integration

The platform can be expanded by integrating additional agricultural data sources. Combining farm records with relevant agricultural datasets can provide a stronger foundation for future AI-based decision-support capabilities.

5.2.7 Future Platform Expansion

The modular architecture of KisanSangam allows additional functionalities and services to be incorporated without significantly changing the existing core architecture. Future development can therefore focus on expanding the platform according to the requirements of farmers and the availability of reliable agricultural data and technologies.

5.3 SUMMARY

The evaluation of KisanSangam demonstrates the integration of its five core functionalities: AI Crop Recommendation, Crop-Specific Farm Guide, Farm Management, Market Price Analysis, and Weather Reporting.

The system combines the introductory website, mobile application, backend services, database, AI/ML component, and external services into a unified agricultural platform. The modular architecture provides a foundation for maintaining the existing functionality and incorporating additional capabilities.

Future development can focus on soil-testing integration, advanced AI and data analytics, multilingual support, cloud scalability, subscription-based services, and integration of additional agricultural data sources.

CHAPTER 6: CONCLUSION AND DISCUSSION

6.1 CONCLUSION

The **KisanSangam – AI-Enabled Smart Agriculture and Farm Management Platform** was developed to provide farmers with a centralized digital platform for accessing important agricultural information and managing farm-related activities.

The system integrates **Artificial Intelligence, mobile application technology, backend services, database management, and external data services** to support agricultural decision-making. The current implementation focuses on five core functionalities: **AI Crop Recommendation, Crop-Specific Farm Guide, Farm Management, Market Price Analysis, and Weather Reporting**.

The **AI Crop Recommendation** module provides crop recommendations based on agricultural parameters provided by the user. The **Crop-Specific Farm Guide** provides relevant information related to the selected crop. The **Farm Management** module enables users to maintain digital records related to their farms, crops, activities, expenses, and production. The **Market Price Analysis** module provides relevant crop market price information, while the **Weather Reporting** module provides current weather and forecast information.

The system also consists of an **introductory web page** that presents information about KisanSangam and a **mobile application** that provides access to the primary agricultural functionalities. The backend serves as the communication layer between the application, database, AI/ML component, and external services.

The system architecture was designed in a modular manner so that the major components can operate independently while communicating through defined interfaces. This approach supports maintainability and provides a foundation for future expansion of the platform.

The evaluation of the implemented system demonstrates the integration of the major modules and their intended workflows. Testing of authentication, crop recommendation, farm management, market price information, and weather reporting provides a basis for verifying the functional behavior of the application.

Although the current implementation focuses on the defined core scope, KisanSangam provides a foundation for further development. Future improvements can include **soil-testing integration, advanced AI and data analytics, multilingual support, cloud scalability, subscription-based services, and integration of additional agricultural data sources.**

In conclusion, KisanSangam demonstrates the application of modern digital technologies and AI in developing an integrated agricultural platform. The system provides a structured foundation for supporting farmers with agricultural information and farm management capabilities while allowing further expansion as the project evolves.

Appendix

A. SOURCE CODE REPOSITORY

The source code of the KisanSangam project is organized according to the different components of the system. The repository contains the application source code, backend services, AI/ML implementation, database-related components, and supporting configuration files.

B. API TESTING

API testing was performed during the development of the KisanSangam system to verify communication between the application and backend services.

The testing covered operations related to:

- User authentication
- Farm management
- Crop records
- Activities
- Expenses
- Production records
- AI crop recommendation
- Market price information
- Weather information

Additional API request and response screenshots can be included in this section if required.

C. SAMPLE DATABASE STRUCTURE

The principal database entities used in KisanSangam include:

Entity	Purpose
Users	Stores user/account information
Farms	Stores farm-related information
Crops	Stores crop records
Activities	Stores farming activity records
Expenses	Stores farm expense records
Production	Stores production-related records

The relationships between these entities are represented in the **Entity-Relationship Diagram (Figure 3.2)** in Chapter 3.

D. AI/ML MODEL SUPPORTING INFORMATION

The AI Crop Recommendation module uses agricultural input parameters to generate a crop prediction.

The general processing flow is:

Input Parameters → Data Preprocessing → Trained ML Model → Prediction → Crop Recommendation

Additional model-related information such as dataset details, preprocessing steps, model evaluation results, and sample predictions may be included here if required.

E. ADDITIONAL SUPPORTING MATERIAL

Additional technical material related to the KisanSangam implementation may be included in this section, such as:

- Additional API responses
- Sample database records
- Additional model outputs
- Deployment configuration
- Technical configuration details
- QR code for the application or repository

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